ATIQUE AHMAD

Python, AI/ML, GenAI, AI Agents, LangGraph, MCP, RAG & LLM Evaluation
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Summary

Python and AI/ML engineer with nearly 4 years of experience building production Python APIs, data pipelines and LLM-powered systems. Specializes in GenAI and agentic applications: LangGraph multi-agent workflows, MCP tool integration, RAG with embedding models, and LLM evaluation for hallucination, faithfulness and agent quality. Fine-tuned a compact 2B model that outperforms GPT-5 on factuality detection, and led the dataset team that built its training data. Works closely with backend, frontend and product teams to turn AI capabilities into reliable, production-ready features.

Key Achievements

- Fine-tuned a **2B-parameter model to 91.7% accuracy**, **+5.4 pts over GPT-5** on the same test set, plus a hallucination detector at **98.1% F1**; both run in production.
- **Led the dataset team** that curated **20,000+ labelled records** powering two production models.
- Built and open-sourced **TRACE**, a full-stack AI security platform with LangGraph agents, MCP tools, human-approved fixes and a live demo.

Experience

AI/ML Engineer & Dataset Team Lead | *Paradigm Networks, Inc.* **Jan 2023 – Present**

LLM Fine-Tuning & Model Development

Built compact in-house models that outperform frontier LLMs on our verification tasks.

- Fine-tuned **IBM Granite Guardian 3.1 (2B)** with **LoRA/PEFT** on **RunPod** GPUs for factuality detection, reaching **91.7% accuracy / 86.7% F1** and outperforming GPT-5 by 5.4 pts on the same test set.
- Trained a hallucination detector to **98.1% F1**; served both models in production with **vLLM** behind async **FastAPI** endpoints.

Dataset Team Lead — Data Curation & Data Engineering

Owned the data side of our in-house AI models, from raw data to training-ready datasets.

- Led a team curating **two datasets of 10,000 records each**, every record pairing a question, source context and answer with a quality label; coordinated labelling and reviewed quality.
- Built Python/Pandas pipelines to clean, validate, deduplicate and merge in-house data with **64K open-source samples** into versioned train/test splits on AWS S3.

Python APIs, Agents & LLM Evaluation

Django REST APIs behind the platform's AI features, with FastAPI serving the models.

- **REST APIs:** built **Django REST Framework** APIs for response verification, answer relevancy and faithfulness, including RAG chunk-level grounding checks (**0.90 F1** with DeepEval vs. 0.85 with RAGAS).
- **LLM providers:** added multi-provider support (OpenAI, Azure OpenAI, AWS Bedrock, GCP Vertex AI) with automatic fallback, and integrated the services with a Go control server and a LangGraph agent platform.
- **AI agents:** developed a **LangGraph code-review agent** that analyses pull-request diffs and posts findings, traced with LangSmith.
- **LLM evaluation:** designed an **agent evaluation service** (FastAPI, MongoDB) that scores every agent run on **9 metrics** (task success, plan quality, hallucination, toxicity), surfaced in the React app.
- **Deployment:** tested with pytest and shipped through Docker, GitHub Actions and **AWS EKS**.

Teaching Assistant / Lab Engineer | *ITU Lahore — Mobile Application Development* **Jan 2024 – Jul 2024**

- Led lab sessions for **100+ students**, taught debugging and app architecture, and guided final-year projects.

Full-Stack Developer Intern | *Clicky* **Jan 2022 – Oct 2022**

- Built and maintained frontend and backend features for production web applications.

Projects

TRACE — AI Security Platform | *LangGraph, MCP, FastAPI, Next.js* Live demo | Open source

- Scans repositories with **Trivy, Semgrep, OSV and Gitleaks**, deduplicates findings and enriches them with CVE intelligence (NVD, GitHub Advisories, CISA KEV).
- Designed a **LangGraph multi-agent workflow** that investigates each finding using **RAG** and permissioned **MCP tools**, citing evidence for every claim.
- Fixes are applied only after **human approval**, then verified by re-scanning the patched code.
- Built an evaluation harness comparing base, RAG and agentic-RAG modes on recall, hallucination and groundedness, plus prompt-injection tests for AI apps.

Retrieval-Augmented Reasoning for Small LLMs | *PyTorch, vLLM, FAISS, Qwen3* Research project

- Built an embedding memory bank and a **600K-edge reasoning graph** with FAISS search; distilled Qwen3-8B reasoning into Qwen3-1.7B, raising held-out accuracy from **75.0% to 85.4%**.

Self-Supervised Learning with SimCLR | *PyTorch, ResNet-18, contrastive learning* Deep learning

- Contrastive pre-training (NT-Xent) on 45K unlabelled CIFAR-10 images reached **80.8% accuracy with 10% of labels**, vs. 74.4% for the supervised baseline.

Visual Similarity Search | *PyTorch, ResNet-50, triplet loss* Deep learning

- Trained triplet-loss embeddings with hard-negative mining for search-by-image across 101 categories; improved Recall@1 from **39.1% to 47.9%**.

Technical Skills

Python & Backend:	Python, Go, SQL, FastAPI, Django REST Framework, async, pytest, REST APIs
AI / LLMs:	LLMs (GPT, Claude, Gemini, Granite, Qwen), embedding models, RAG, LangChain, LangGraph, LangSmith, MCP, DeepEval, fine-tuning (LoRA/PEFT), vLLM, Hugging Face, PyTorch, deep learning (Transformers, RNNs, LSTMs, GANs)
Data:	Pandas, NumPy, ETL pipelines, PostgreSQL, MongoDB, Redis, FAISS, Qdrant
Cloud & DevOps:	AWS (S3, EKS, Bedrock), RunPod (GPU training), GCP Vertex AI, Azure OpenAI, Docker, Kubernetes, GitHub Actions
Web:	TypeScript, React, Next.js
Security:	Semgrep, Trivy, Gitleaks, CVE intelligence, prompt-injection testing

Education

M.S. Data Science | *Information Technology University, Lahore* **2025 – 2027**

B.S. Computer Engineering | *Information Technology University, Lahore* **2019 – 2023**